Milestone 3

Group 9

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We compare frozen Stable Diffusion 1.5 with rank-4 LoRA on 500 matched Flickr30k test prompts. LoRA lowers FID from 112.283 to 106.999, raises Inception Score from 13.291 to 14.091, and raises CLIP alignment from 32.368 to 32.982. Visual inspection nevertheless reveals errors in counts, poses, and text. Validation experiments separately examine guidance scale and text embedding choice.

1 Methods

Data preparation

From a pinned lmms-lab-encoder/flickr30k revision, we sample 5,500 candidate images using seed 42 and a streaming shuffle buffer of 2,048. Processing normalizes captions, removes exact duplicates, corrects orientation, and converts images to RGB. We retain 5,000 pairs with one caption per image and center-crop to 512 × 512 pixels. A custom seeded image-level split separates training and evaluation. The source split named test is a storage container; these partitions do not reproduce the standard Flickr30k benchmark.

Table 1. Dataset partitions and evaluation roles.

Partition	Pairs	Use
Training	4,000	LoRA optimization
Validation	500	8 fixed validation samples for checkpoint selection; 100 fixed validation prompts for parameter comparisons.
Test	500	Final matched model comparison

Model adaptation and optimization

Stable Diffusion 1.5 uses VAE latents and CLIP text conditioning. We add rank-4 LoRA adapters to the query, key, value, and output projections of the attention layers. We train only these adapters while keeping the original model weights frozen. Optimization uses noise-prediction MSE, AdamW at learning rate 0.0001 and weight decay 0.01, gradient clipping at 1.0, and caption dropout of 0.1.

We run 1,000 updates with batch size 1 and four accumulation steps. Across 1,000 updates, the model processes four images per update, totaling 4,000 image presentations. Because images are randomly sampled at each update, some may be used multiple times while others may never be selected. Checkpoints are evaluated every 100 updates on eight fixed validation cases. Step 1,000 achieves the lowest recorded validation MSE, 0.08718, and is used for testing.

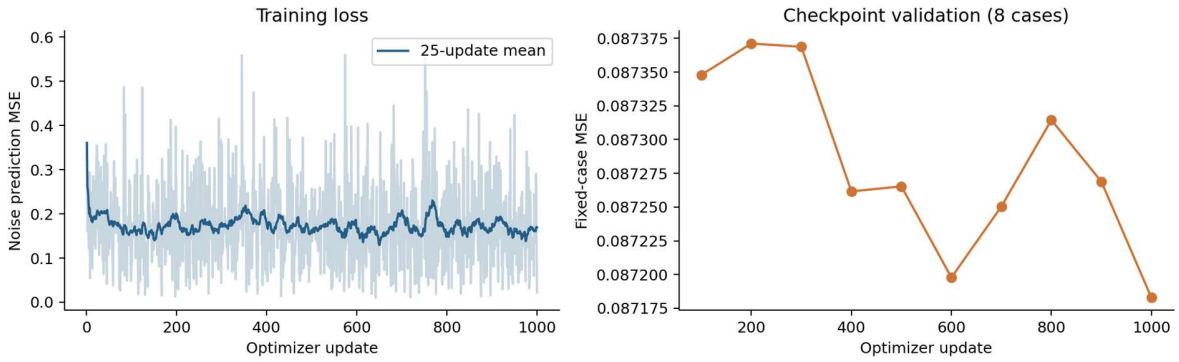


Figure 1. Training MSE with a 25-update moving mean and checkpoint loss on eight fixed validation cases.

Evaluation protocol

Both test models use DDIM with leading timestep spacing, 30 sampling steps, guidance scale 7.5, and the final CLIP hidden layer. Each caption receives seed 42 plus its index, yielding matched initial noise across the two models. Images are generated at 512 × 512 pixels with an empty negative prompt. The pretrained safety checker remains enabled. Its saved replacement images remain in the metric calculations, so the evaluation describes the complete generation pipeline.

Table 2. Controlled experimental design.

Experiment	Configurations	Images per setting	Model
Guidance scale	1, 3, 5, 7.5, 10, 12	100	Frozen SD 1.5
Text conditioning	Final layer; penultimate layer plus LayerNorm	100	Frozen SD 1.5
Additional ablations	20 or 50 steps; Euler; DPM; trailing spacing	100	Frozen SD 1.5
Final test	Fixed baseline inference settings	500	Frozen SD 1.5 and LoRA

The validation sweep contains 12 distinct settings, including the shared baseline. Each changes one factor while keeping the validation captions and seeds fixed. Classifier-free guidance amplifies the difference between conditional and unconditional denoising predictions. Scale 1 uses conditional generation without this amplification. The embedding ablation takes the penultimate CLIP hidden state and applies the same final LayerNorm. Both models are evaluated using the same fixed test settings. The validation sweep analyzes parameter sensitivity but does not automatically determine the final test configuration.

Quantitative metrics

Fréchet Inception Distance (FID) compares Gaussian approximations to real and generated image feature distributions. It combines the squared distance between their feature means with a covariance discrepancy. Lower values indicate closer distributions in the chosen feature space. For FID, we extract 2,048-dimensional Inception features, accumulate feature statistics in float64, and compare equally sized sets of real and generated images.

Inception Score (IS) evaluates generated images using their predicted class probabilities. It is the exponential of the expected Kullback–Leibler divergence between each image's predicted class distribution and the marginal class distribution across the generated set. Higher values favor confident predictions across varied classes. We report the mean and standard deviation over 10 splits: 10 images per split for validation and 50 per split for testing. This standard deviation is not a confidence interval, and IS does not measure caption agreement.

FID and IS use TorchMetrics 1.8.1 and torch-fidelity 0.3.0. Real and generated images are RGB uint8 tensors, with the same explicit bilinear resize to 299 × 299 pixels. Consistent preprocessing is essential for interpreting model differences. FID values from the 100-image validation sweep and the 500-image test comparison are treated separately.

CLIP alignment supplements these metrics with prompt-specific similarity. We use the separate openai/clip-vit-base-patch32 evaluator and report 100 times the nonnegative cosine similarity of normalized text and image embeddings. This evaluator stays fixed when the generation embedding changes. Its score measures broad semantic agreement but does not guarantee correct counting, spatial relations, or spelling.

Paired analysis and qualitative review

We compute the LoRA-minus-baseline CLIP change for each test caption. A paired bootstrap resamples 500 captions with replacement 2,000 times and uses the 2.5th and 97.5th percentiles of the mean change. This interval is conditional on the saved generation seeds; it does not measure variation across retraining runs. No confidence intervals are estimated for FID or IS differences.

Qualitative inspection examines the first four test cases in fixed order, followed by explicitly score-selected diagnostic cases. We assess requested objects, counts, attributes, actions, relations, and visible artifacts. Reference photos provide semantic context, not a reconstruction target.

2 Results

Quantitative model comparison

Table 3. Held-out test results with 500 generated images per model.

Model	FID	IS mean ± SD	CLIP	Seconds per image
Frozen SD 1.5	112.283	13.291 ± 1.365	32.368	5.097
SD 1.5 with LoRA	106.999	14.091 ± 1.360	32.982	5.344

LoRA improves all three aggregate quality metrics in this run. FID falls by 5.284 points (4.71%), IS rises by 0.801 (6.02%), and mean CLIP alignment rises by 0.614 (1.90%). These are descriptive differences from one training run and one saved seed assignment. The IS error bars describe variation across score splits, so their overlap does not establish a statistical test of the model difference.

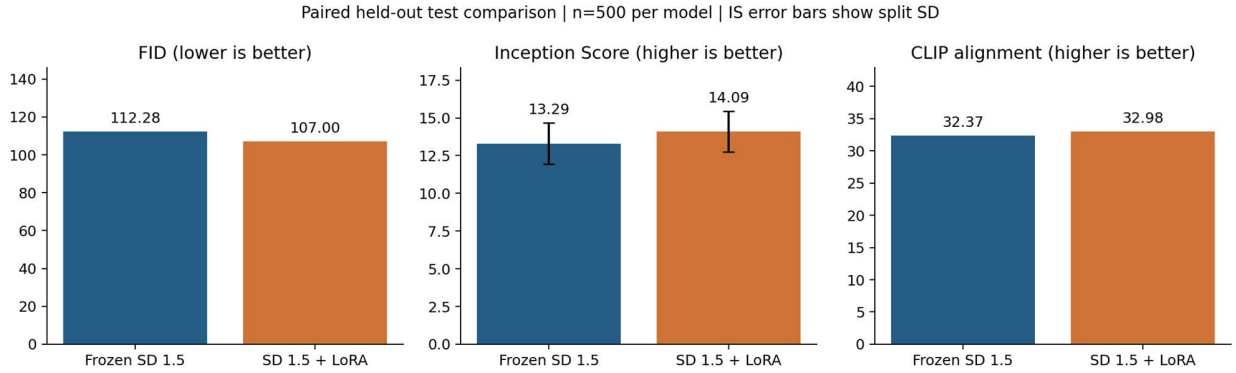


Figure 2. Matched test comparison. Lower FID is better; higher IS and CLIP are better. IS error bars show split standard deviation.

Table 4. Paired CLIP changes across the same 500 captions.

Mean change	Median change	95% bootstrap interval	Prompts with higher LoRA score
+0.614	+0.548	[+0.406, +0.831]	303 / 500 (60.6%)

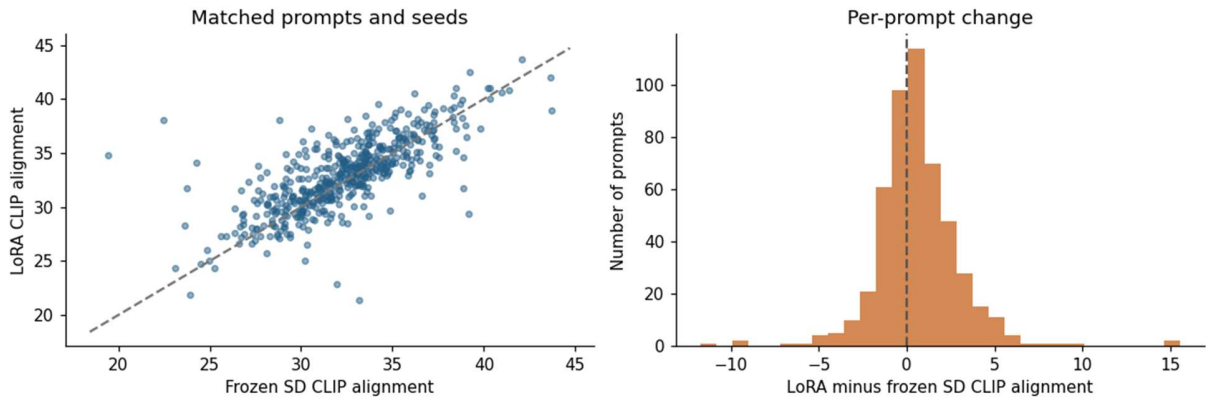


Figure 3. Per-caption CLIP scores and paired differences. Points above the diagonal favor LoRA.

The positive bootstrap interval supports a higher average CLIP score for this caption set under the fixed seed assignment. It does not imply improvement for every prompt. Safety flags occur in 3 baseline outputs (0.6%) and 2 LoRA outputs (0.4%). Their inclusion matters because a black replacement image can create an extreme paired score change. Recorded generation time increases by 0.247 seconds per image (4.84%); these logs are not a controlled hardware benchmark.

Guidance scale sensitivity

Table 5 isolates guidance scale using the same 100 validation prompts, seeds, and frozen model. DDIM, 30 steps, leading spacing, and final-layer CLIP conditioning remain fixed. The strongest change occurs between guidance 1 and 3: FID drops by 28.622 points and CLIP rises by 4.026. Increasing guidance from 1 to 7.5 raises CLIP by 4.901 points overall.

Table 5. Guidance sensitivity on 100 validation images per setting.

Guidance	FID	IS mean ± SD	CLIP	CLIP change from 7.5
1.0	251.256	4.872 ± 0.600	27.351	-4.901
3.0	222.635	5.599 ± 1.063	31.377	-0.875
5.0	222.955	5.616 ± 1.022	31.707	-0.544
7.5	227.689	5.795 ± 1.130	32.251	+0.000
10.0	227.755	6.084 ± 1.155	32.204	-0.047
12.0	226.704	6.032 ± 1.204	32.298	+0.047

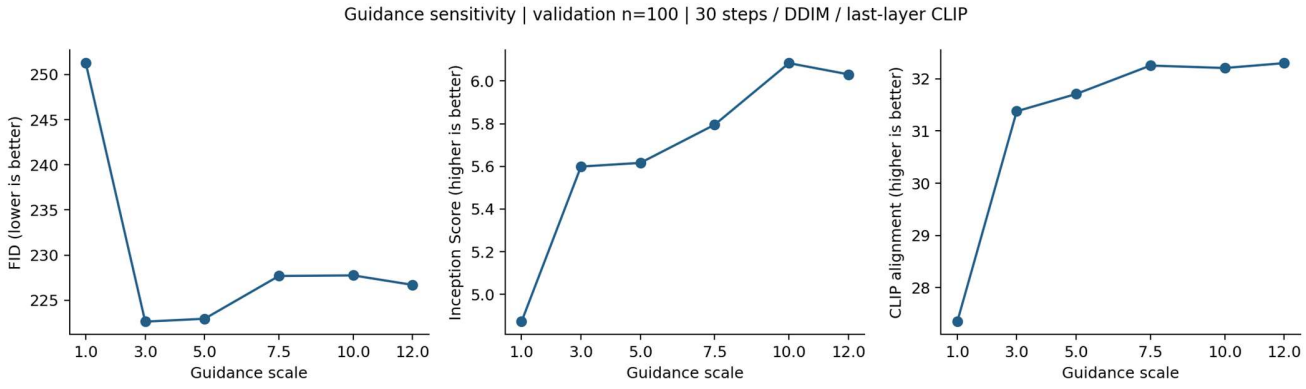


Figure 4. One-factor guidance sweep. The same images and sample count are used for every validation setting.

No scale is best on every metric. Guidance 3 has the lowest FID, 222.635. Guidance 10 has the highest IS, 6.084, among the guidance settings. Guidance 12 has the highest mean CLIP score, 32.298, but exceeds guidance 7.5 by only 0.047. Its paired 95% bootstrap interval relative to 7.5 is [−0.484, +0.490], which includes zero. Guidance 10 also shows no clear average CLIP advantage over 7.5, with a change of −0.047 and interval [−0.555, +0.374].

These results suggest that guidance substantially improves alignment over scale 1, while average alignment gains flatten around 7.5 in this sample. The FID and IS rankings continue to vary, so higher guidance does not provide a monotonic quality improvement. The validation sweep supports a tradeoff between criteria, rather than a single universally optimal value. Guidance 7.5 remains the fixed test setting; it is not presented as a value selected after observing the test results.

At 100 images per setting, the feature covariance estimates are based on a small sample relative to the 2,048-dimensional feature space. The sweep rankings are exploratory.

Embedding choice and additional ablations

Replacing the final CLIP hidden layer with the penultimate layer plus final LayerNorm lowers validation FID by 2.833 points, but also lowers IS by 0.225 and mean CLIP alignment by 0.817. Thus, a closer marginal image distribution does not necessarily imply better agreement with the requested caption.

Table 6. Embedding comparison with guidance 7.5 and 30 DDIM steps.

Conditioning	FID	IS mean ± SD	CLIP
Final CLIP layer	227.689	5.795 ± 1.130	32.251
Penultimate layer plus LN	224.857	5.571 ± 0.929	31.434

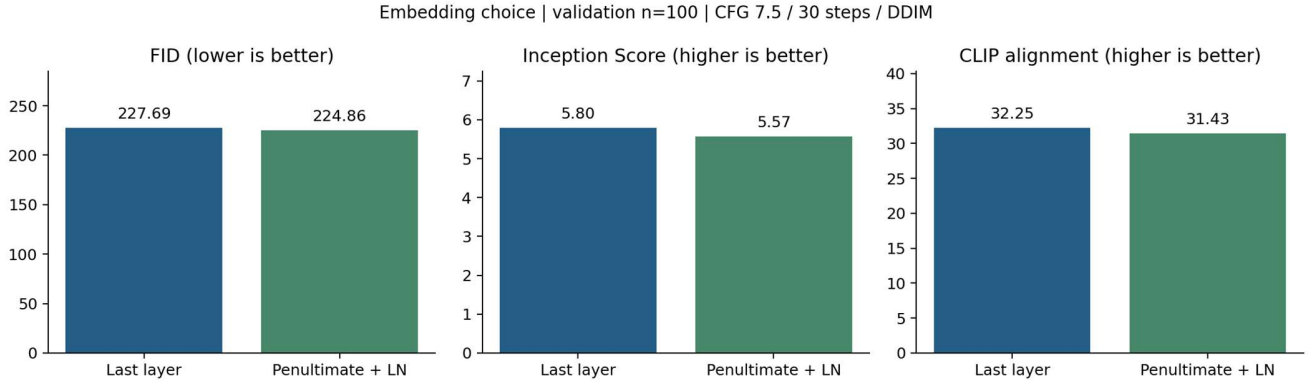


Figure 5. Text embedding ablation on 100 validation prompts. LN denotes the final LayerNorm in the same text encoder.

The paired CLIP change is −0.817, with a conditional 95% bootstrap interval of [−1.286, −0.330]; only 37% of prompts improve. This favors final-layer conditioning for average caption alignment in this run. It does not establish that final-layer embeddings are superior for every prompt or that replacing the entire text encoder would have the same effect.

Table 7. Additional one-factor validation comparisons with n = 100.

Setting	FID	IS mean ± SD	CLIP	Seconds per image
Baseline	227.689	5.795 ± 1.130	32.251	5.067
20 sampling steps	227.405	6.028 ± 1.269	32.052	3.463
50 sampling steps	225.378	6.129 ± 1.274	32.008	8.318
Euler sampler	228.054	5.752 ± 1.114	32.269	5.081
DPM multistep sampler	225.337	5.727 ± 1.052	32.173	5.078
Trailing timestep spacing	227.226	5.804 ± 1.116	31.840	5.084

Increasing sampling from 20 to 50 steps raises logged time from 3.463 to 8.318 seconds per image without increasing mean CLIP alignment. The 50-step setting has the highest IS in the full sweep, 6.129. Euler's CLIP change relative to the baseline is only +0.018, while trailing spacing lowers it by 0.411. These comparisons reinforce that metric choice and computational cost affect how a configuration is judged.

Prompt and image alignment in fixed examples

The first two test cases illustrate why aggregate alignment scores require visual interpretation. Captions, reference photos, and generated images retain their notebook order. Scores below use the same CLIP evaluator as the aggregate analysis.



Figure 6. First two test prompts. Columns show the reference photo, frozen SD 1.5, and LoRA.

Picnic scene

Prompt: “Three people having a picnic under two trees.” Both models clearly depict a picnic beneath trees, with three prominent people and an additional partially visible person at the right edge. The exact request for three people is therefore not cleanly satisfied, and the tree count is ambiguous. LoRA changes the composition slightly and raises CLIP from 30.277 to 31.186, but the higher score does not resolve the counting issue.

People on a mountain summit

Prompt: “One person stands and another sits on the summit of a mountain.” The baseline shows one seated person. LoRA shows two people, improving the count, but both appear seated. CLIP rises from 30.055 to 31.590. This is a partial semantic improvement: the requested standing-versus-sitting relation remains incorrect. The center-cropped reference itself has no clearly visible people, so caption adherence should be judged against the prompt rather than exact visual similarity to that crop.

Actions objects and visual structure

The next two fixed-order cases compare action recognition and object structure. Their CLIP differences are small, and the visual evidence does not support a clear overall winner.



Figure 7. Third and fourth test prompts in fixed order.

Reference photos are semantic examples rather than target reconstructions.

Airborne bicycle trick

Prompt: “Biker doing a trick in air.” Both models depict a rider performing an airborne stunt. However, the wheel, frame, and body geometry are implausible; LoRA introduces conspicuous overlapping circular structures near the upper body. CLIP changes only from 33.137 to 33.014. Both outputs capture the broad action while retaining structural artifacts that a semantic similarity score does not adequately summarize.

Soccer in the rain

Prompt: “two boys play soccer in the rain.” Both outputs contain two children, a ball, visible rain, and a wet playing surface. LoRA changes the limb positions and ball placement; the ball appears suspended beside a hand, which makes the soccer action less explicit. CLIP is nearly unchanged, 37.241 versus 37.211. The visual comparison supports similar broad alignment, with remaining ambiguity in the action and object relationships.

Diagnostic examples and score extremes

The following examples are deliberately selected by CLIP scores. They identify failure modes and should not be interpreted as representative samples or evidence of a population-wide success rate.

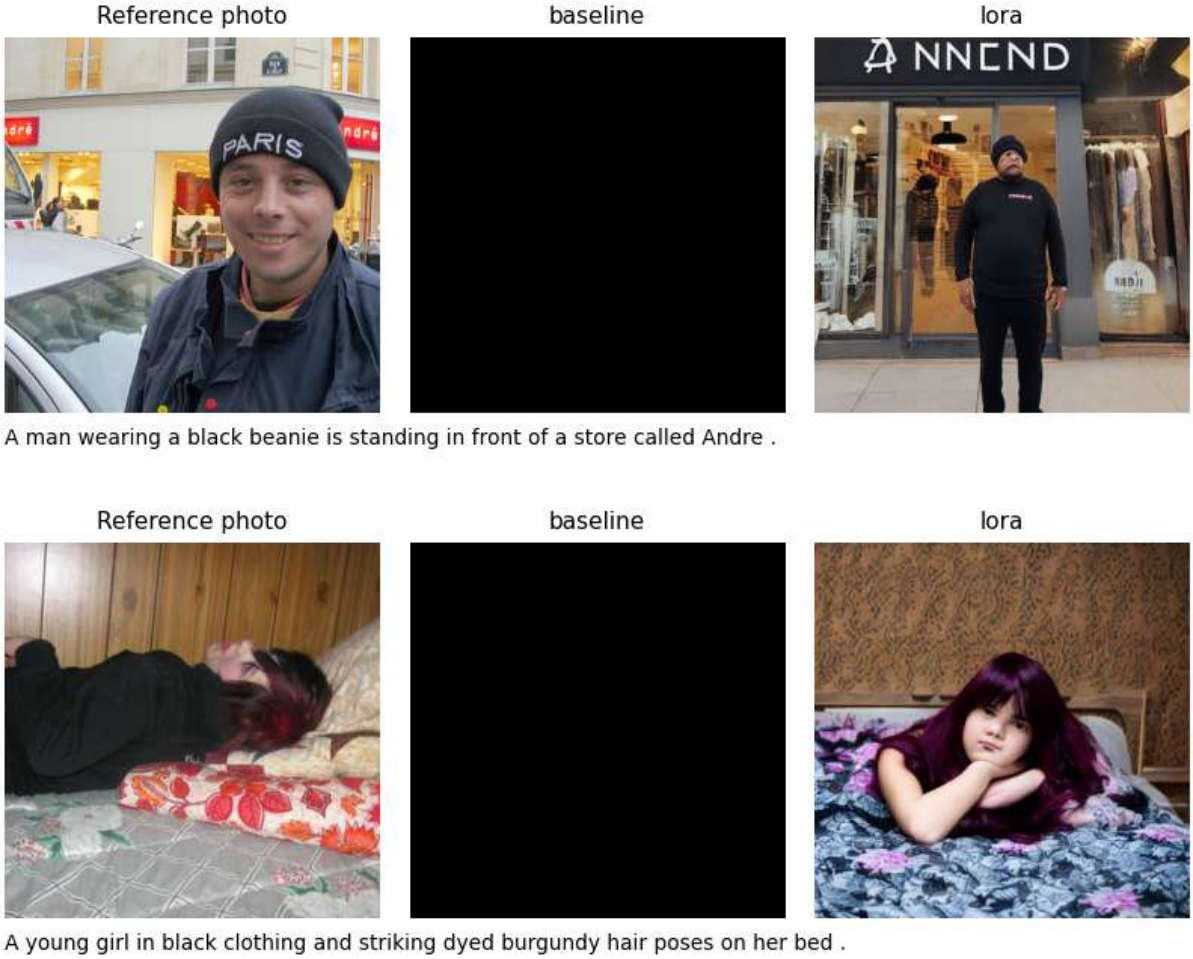


Figure 8. The largest positive CLIP change and the lowest baseline CLIP case.

The baseline outputs are black; the LoRA outputs recover visible prompt content.

Storefront and text rendering

The prompt asks for a man in a black beanie standing in front of a store called Andre. The saved baseline image is black. LoRA produces the man, beanie, and storefront, raising CLIP from 22.465 to 38.051 (a gain of 15.586). The sign does not correctly spell Andre. The score gain therefore reflects recovery of broad content while exact text alignment still fails.

Hair color clothing and safety replacements

The lowest-baseline-score prompt describes a girl in black clothing with burgundy hair posing on a bed. The baseline is black, whereas LoRA depicts the requested hair color, dark clothing, and bed setting. CLIP rises from 19.441 to 34.816. All saved images remain in the reported metric calculations. These extreme cases reflect the delivered pipeline, including its enabled safety checker, and are not clean comparisons of two unobstructed scenes.

Visual parameter comparison and limitations



Figure 9. The first two validation prompts across guidance 1, 3, 7.5, and 12, followed by penultimate-layer conditioning at guidance 7.5.

All comparisons use the same prompt-specific seed.

For the microscopy prompt, guidance 1 produces a crowded scene with weak object definition. Stronger guidance produces clearer children and equipment-like objects, but neither microscope structure nor adult supervision is consistently unambiguous. Penultimate-layer conditioning changes the composition without clearly solving these requirements. This visual pattern is consistent with the strong score gain over guidance 1 and the much smaller gains at higher scales.

For the café prompt, guidance 7.5 and 12 emphasize an elderly couple, while the requested woman working on a MacBook is absent. Penultimate-layer conditioning emphasizes an older man using a laptop, losing the couple and the requested woman. Both outputs capture portions of the caption, but distributing the correct actions and objects across multiple people remains difficult. These two cases illustrate the failure mode; they do not estimate its frequency.

Interpretation and limitations

The comparison on 500 held-out test prompts shows that the LoRA adapter improves all three aggregate metrics in this experiment. FID decreases from 112.283 to 106.999, Inception Score increases from 13.291 to 14.091, and mean CLIP alignment increases from 32.368 to 32.982. These results suggest that the generated image distribution becomes closer to the reference distribution in Inception feature space, while average text–image similarity also improves. However, these gains describe one experimental run. They do not establish that LoRA consistently improves every image or will produce the same benefits under different training and generation seeds.

The visual examples help explain the limits of these numerical improvements. Such images can receive relatively high CLIP scores because they contain the main subjects and setting described by the prompt. Exact counts, body structure, written text, and relationships between people can still be incorrect. Therefore, the increase in mean CLIP alignment should be interpreted as improved average semantic similarity, rather than proof of complete instruction following.

The evaluation sample sizes also limit the strength of the conclusions. Each validation configuration contains 100 generated images, whereas each final test model is evaluated on 500 images. FID estimates the means and covariances of 2,048-dimensional image features, so estimates based on these relatively small samples may be sensitive to which images are included. Inception Score is also affected by the evaluation procedure: both partitions use ten splits, giving

only ten images per split for validation but fifty per split for testing. Because the image subsets, sample counts, and split sizes differ, validation and test scores should be interpreted within their respective comparisons. For example, a higher test IS than validation IS does not by itself demonstrate that the test configuration is better. Moreover, we do not estimate confidence intervals for the FID or IS differences, so their observed improvements should not be presented as statistically established effects.

Randomness is another source of uncertainty. The experiment uses one LoRA training seed and one generation seed for each caption. The baseline and LoRA models share the same caption-specific seed, which makes their comparison more controlled, but it does not show how either model behaves across alternative seeds. The paired CLIP bootstrap resamples the existing caption–image results; it does not retrain the model or generate new images. Its 95% interval of approximately [0.406, 0.831] therefore supports a positive average CLIP change conditional on the saved outputs. It does not measure variability across independent training runs or alternative generations of the same caption. Repeated training and generation would be necessary to assess whether the observed advantage is reproducible.

Checkpoint selection provides a further limitation. The selected checkpoint, at update 1,000, has the lowest recorded denoising loss on eight fixed validation cases. Using the same cases makes checkpoint losses comparable, but eight cases cover only a small part of the validation set. This result identifies the best checkpoint under that particular selection procedure; it does not demonstrate convergence across the full validation distribution. Denoising loss also measures a different objective from image realism or caption adherence, so the checkpoint with the lowest validation loss is not necessarily the checkpoint with the best FID, IS, or visual quality.

Data preparation may introduce additional uncertainty. Removing exact duplicate images reduces direct overlap between the custom training, validation, and test partitions. However, it does not identify every near duplicate, such as a different crop of the same photograph, or establish whether evaluation images appeared in the base model's pretraining data. Center cropping can also remove objects or people mentioned in the original caption.

Finally, the metrics evaluate the complete generation pipeline, including its safety checker. Three baseline outputs and two LoRA outputs are flagged, and saved replacement images remain in the calculations. The diagnostic examples show that black outputs can contribute to large positive or negative CLIP changes. Consequently, some differences reflect the interaction between generation and filtering, rather than differences between two visible generated scenes.

https://github.com/Sneha0504/IE7615_Generative_Project